Yeast Datasets Readout Table

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✓ read and checked ■ read, provisional ✗ agent-written, not yet read

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Table 1. The 70 *S. cerevisiae* datasets the served knowledge graph carries, grouped by readout. *Experiments* is the dataset's experiment count in the release snapshot. *Gene-level* is yes when the readout is a property of a gene or a named locus (an expression level, a protein abundance, a copy number) and no when it is a property of the whole cell (growth, viability, morphology, a metabolite or product level). *New* marks a dataset absent from the previous release. Release `2026.10.17-4295f7ca` (KG 4.0, torchcell v1.8.3, built 2026-10-17, 143,075,580 nodes over 3 organisms), compared with `2026.10.06-4b293d34` (KG 3.0). Generated 2026-10-18 01:00 CDT by `experiments/database/scripts/yeast_datasets_readout_table.py`; the PDF is `notes/assets/pdf-output/experiments.database.yeast-datasets-readout-table.pdf`.

1.1 Expression ✗

Dataset	Experiments	Gene-level	Readout	Genotype	New
Boocock2025	79,516	yes	single-cell expression	segregants, four crosses	new
PooledPromoterExpressionVaishnav2022	71,158	yes	reporter expression per promoter, sort-seq	random promoter library	new
NadalRibellesPerturbSeq2025	6,188	yes	pseudobulk expression, Perturb-seq	genome-scale deletions	
Nguessan2025	2,951	yes	single-cell expression	segregants, BY x RM	new
InductionTimecourseHackett2020	1,752	yes	expression timecourse, microarray	inducible regulator overexpression	new
MicroarrayKemmeren2014	1,484	yes	expression, microarray	kanMX deletions	
CaudalPanTranscriptome2024	943	yes	expression, RNA-seq TPM	natural isolates	
TfDeletionPseudobulkJackson2020	784	yes	pseudobulk single-cell expression	TF deletions	new
MicroarrayScoreHu2007	263	yes	expression score, microarray	TF deletions	new
SmMicroarraySameith2015	82	yes	expression, microarray	GSTF single deletions	
DmMicroarraySameith2015	72	yes	expression, microarray	GSTF double deletions	

1.2 Proteome ✗

Dataset	Experiments	Gene-level	Readout	Genotype	New
ProteomeMessner2023	4,699	yes	proteome	kanMX deletions	
ProteomeMuenzner2024	795	yes	proteome, SWATH-MS	natural isolates	new
ProteomeZelezniak2018	97	yes	proteome	kinase deletions	
ProteomeMessner2021	70	yes	proteome	BY4741 under 16 drugs	new
ProteinTurnoverMuenzner2024	55	yes	protein turnover, dynamic SILAC	natural isolates	new

1.3 Genome architecture ✗

Dataset	Experiments	Gene-level	Readout	Genotype	New
GenomeArchitecturePuddu2019	8,835	yes	copy number of named loci	YKO deletions, homozygous diploid	new

1.4 Fitness and genetic interaction ✗

Dataset	Experiments	Gene-level	Readout	Genotype	New
DmfCostanzo2016	20,705,612	no	fitness	SGA double mutants	
DmiCostanzo2016	20,705,612	no	genetic interaction	SGA double mutants	
DmfKuzmin2020	632,998	no	fitness	SGA double mutants	
DmiKuzmin2020	632,797	no	genetic interaction	SGA double mutants	
DmfKuzmin2018	410,571	no	fitness	SGA double mutants	
DmiKuzmin2018	410,399	no	genetic interaction	SGA double mutants	
TmfKuzmin2020	301,798	no	fitness	SGA triple mutants	
TmiKuzmin2020	301,798	no	genetic interaction	SGA triple mutants	
TmfKuzmin2018	91,111	no	fitness	SGA triple mutants	
TmiKuzmin2018	91,111	no	genetic interaction	SGA triple mutants	
SmfCostanzo2016	20,484	no	fitness	SGA single mutants	
SmfBaryshnikova2010	6,023	no	fitness	SGA single mutants	
SmfKuzmin2018	1,539	no	fitness	SGA single mutants	
SmfODuibhir2014	1,312	no	fitness, relative growth rate	kanMX deletions	
SmfKuzmin2020	472	no	fitness	SGA single mutants	

1.5 Viability ✗

Dataset	Experiments	Gene-level	Readout	Genotype	New
SynthLethalityYeastSynthLethDb	14,000	no	synthetic lethality	gene pairs	
SynthRescueYeastSynthLethDb	6,948	no	synthetic rescue	gene pairs	
GeneEssentialitySgd	1,329	no	essentiality, viability	deletions and alleles	

1.6 Environment and chemical genomics ✗

Dataset	Experiments	Gene-level	Readout	Genotype	New
EnvChemgenPiotrowski2017	3,430,343	no	growth response z-score, barcode pool	barcoded deletion pool plus compounds	new
EnvChemgenHoepfner2014	3,127,814	no	HIP-HOP sensitivity score	barcoded deletions, het and hom	
HetHillenmeyer2008	2,963,683	no	fitness	heterozygous deletions plus compounds	
HomHillenmeyer2008	1,182,916	no	fitness	homozygous deletions plus compounds	
EnvChemgenWildenhain2015	448,655	no	growth response z-score	barcoded deletions plus compounds	
EnvChemgenParsons2006	300,834	no	growth response, barcode pool	barcoded deletions plus compounds	new
CrisprMagicLian2019	266,304	no	guide enrichment	CRISPRa, CRISPRi, CRISPRd plus furfural	
EnvChemgenVanacloig2022	164,205	no	growth response	barcoded deletions plus compounds	
EnvChemgenCostanzo2021	61,640	no	condition fitness	SGA single mutants	
EnvChemgenYoshikawa2009	37,832	no	growth response, barcode pool	barcoded deletions plus compounds	new
FattyAcidSmith2006	14,310	no	clear-zone score, ordinal	kanMX deletions	
CrispriChemgenSmith2016	7,053	no	guide fitness	CRISPRi plus drugs	
EnvChemgenMota2024	1,287	no	growth response	kanMX deletions	
InhibitorBioscreenVolk2021	977	no	Bioscreen growth (private)	in-house strain under inhibitors	new
EnvChemgenAuesukaree2009	525	no	stress tolerance, categorical	kanMX deletions	
CrispriMormino2022	12	no	growth response	CRISPRi plus acetic acid	

1.7 Natural isolates and segregants ✗

Dataset	Experiments	Gene-level	Readout	Genotype	New
CrispriSegregantHale2024	8,692,635	no	growth response	segregants plus CRISPRi	new
Bloom2019	530,100	no	growth	segregants, 16 crosses	
DoublingTimeDeChiara2022	36,295	no	doubling time	natural isolates	new
CellYieldDeChiara2022	36,286	no	cell yield	natural isolates	new
IsolateGrowthPeter2018	33,985	no	colony growth	natural isolates	new
LifeCycleDeChiara2022	4,039	no	sporulation and life span	natural isolates	new

1.8 Morphology ✗

Dataset	Experiments	Gene-level	Readout	Genotype	New
ScmdOhya2005	4,718	no	morphology, CalMorph	kanMX deletions	
ScmdOhnuki2022	1,982	no	morphology, CalMorph	drug-hypersensitive deletions	
ScmdOhnuki2018	1,112	no	morphology, CalMorph	essential-gene heterozygous diploids	

1.9 Metabolite and product ✗

Dataset	Experiments	Gene-level	Readout	Genotype	New
IsobutanolScreenLopez2024	4,805	no	biosensor fluorescence	kanMX deletions	
BetaxanthinCachera2023	4,759	no	betaxanthin proxy	CRI-SPA deletions and gene additions	
AminoAcidMulleder2016	4,678	no	amino-acid levels	kanMX deletions	
CarotenoidOzaydin2013	4,607	no	colony color score	deletions plus carotenoid pathway	
AminoAcidCooper2010	4,382	no	amino-acid levels	kanMX deletions	
IsobutanolValidatedLopez2024	227	no	isobutanol titer	kanMX deletions	
FattyAcidXue2025	176	no	free fatty acid titer (in-house)	combinatorial deletions	
MetaboliteZelezniak2018	129	no	metabolome	kinase deletions	
MetaboliteDaSilveira2014	127	no	lipidome	kinase and phosphatase deletions	
OrganicAcidYoshida2012	17	no	organic-acid titers	kanMX deletions	

1.10 Totals ✗

70 datasets, 65,889,107 experiments; 17 gene-level, 53 whole-cell; 19 new since `2026.10.06-4b293d34`, and 18 already-served datasets gained records under the store-what-was-released loader fixes (HetHillenmeyer2008 2,712,677 to 2,963,683, HomHillenmeyer2008 1,063,034 to 1,182,916, EnvChemgenVanacloig2022 118,662 to 164,205, EnvChemgenHoepfner2014 3,083,827 to 3,127,814, and 14 others).

Data: `experiments/database/results/yeast_datasets_readout_table.csv`. Script note: `experiments.database.scripts.yeast_datasets_readout_table`.